

WASP-12b

R-1502

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-12_190109 · created 2026-08-01T14:57:16+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2458492.65942 +/- 0.00057 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1214 +/- 0.0099
Transit depth (Rp/Rs)^2	1.47 +/- 0.24 [%]
Orbital Inclination (inc)	77.28 +/- 2.77
Airmass coefficient 1 (a1)	1.0278 +/- 0.0084
Airmass coefficient 2 (a2)	-0.0254 +/- 0.0029
Scatter in the residuals of the lightcurve fit is	0.81 %
Best Comparison Star	#1 - [267, 180]
Optimal Aperture	5.02
Optimal Annulus	9.21
Transit Duration (day)	0.105 +/- 0.013

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.1214 ± 0.0099 vs 0.1178 (deviation 0.35σ)
Duration fit vs published	0.105 d vs 0.125 d (deviation 1.47σ)
Mid-time O–C	1.0 min
Depth SNR	11.7
Residual scatter	0.81 %

Light curve

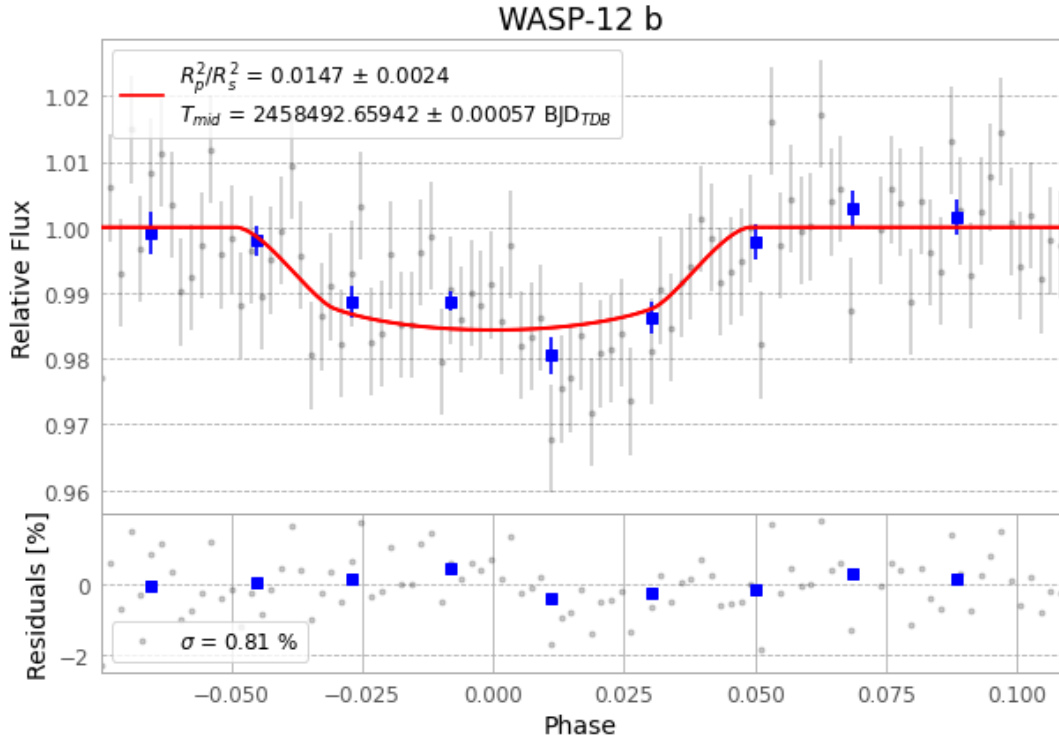


Figure 1 — FinalLightCurve_WASP-12 b_2019-01-08.png (SHA-256 571016c9cc277c65...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [346, 143], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 fbddfc8bc820b2d5...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

97 FITS frames (64 MB), WASP-12190109014212.FITS → WASP-12190109063012.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-190109.log (713988fe32ed...), FinalLightCurve_WASP-12 b_2019-01-08.png (571016c9cc27...), FinalLightCurve_WASP-12 b_2019-01-08.csv (8d156bcd0bea...)

Compute resources

Wall time 145.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 14:57Z.